Rovnice v součinovém tvaru

$A \cdot B = 0 \Leftrightarrow A = 0 \vee B = 0$

1. Řešte rovnice

a) $x(x+3) = 0$

$x_1 = 0$ → $x + 3 = 0$

$x_2 = -3$

$K = \{0; -3\}$

b) $(3x-1)\left(\frac{x}{2}+4\right) = 0$

$3x - 1 = 0$ $\quad \frac{x}{2} + 4 = 0$

$x_1 = \frac{1}{3}$ $\quad x_2 = -8$ $\quad K = \{\frac{1}{3}; -8\}$

c) $(4-2y)(-3y-4) = 0$

$4 - 2y = 0$ $\quad -3y - 4 = 0$

$y = 2$ $\quad y = -\frac{4}{3}$ $\quad K = \{2; -\frac{4}{3}\}$

2. Řešte rovnice

a) $3x^2 - 1 = x - 1$ $\quad / -x + 1$

$3x^2 - x = 0$

$x(3x - 1) = 0$

$x_1 = 0$ $\quad x_2 = \frac{1}{3}$ $\quad K = \{0; \frac{1}{3}\}$

c) $y\sqrt{2} = 2y^2$

$y\sqrt{2} - 2y^2 = 0$

$y(\sqrt{2} - 2y) = 0$

$y_1 = 0$ $\quad \sqrt{2} - 2y = 0$

$\sqrt{2} = 2y$

$y_2 = \frac{\sqrt{2}}{2}$ $\quad K = \{0; \frac{\sqrt{2}}{2}\}$

d) $28 - 7x^2 = 0$ $|:4$

$4 - x^2 = 0$

$(2-x)(2+x) = 0$

$x_1 = 2 \quad x_2 = -2$ $\quad K = \{-2; 2\}$

3. Řešte rovnice

a) $x^2 + x(x+1) = 0$

$x^2 + x(x+1) = x \cdot x + x(x+1) =$

$x(x + x + 1)$

$x(x + x + 1) = 0$

$x(2x + 1) = 0$

$x_1 = 0 \quad x_2 = -\frac{1}{2} \quad K = \{0; -\frac{1}{2}\}$

b) $(x+2)x = (2x-1)(x+2)$

NIKDY nedělím neznámou x!

$(x+2)x - (2x-1)(x+2) = 0$

$(x+2)(x - 2x + 1) = 0$

$(x+2)(-x+1) = 0$

$x_1 = -2 \quad x_2 = 1$

$K = \{-2; 1\}$

4. Řešte rovnice

a) $4x^3 - x = 0$

$x(4x^2 - 1) = 0$

$x(2x-1)(2x+1) = 0$

$x_1 = 0 \; ; \; x_2 = \frac{1}{2} \; ; \; x_3 = -\frac{1}{2}$ $\quad K = \{0; \frac{1}{2}; -\frac{1}{2}\}$

b) $3t^3 = \frac{1}{3}t$

$3t^3 - \frac{1}{3}t = 0 \; | \cdot 3$

$9t^3 - t = 0$

$t(9t^2 - 1) = 0$

$t(3t-1)(3t+1) = 0$

$t_1 = 0$

$t_2 = \frac{1}{3}$

$t_3 = -\frac{1}{3}$

$K = \{0; \frac{1}{3}; -\frac{1}{3}\}$

c) $x^3 + 2x^2 - x - 2 = 0$

$x^2(x+2) - (x+2) = 0$

$(x+2)(x^2-1) = 0$

$(x+2)(x-1)(x+1) = 0$

$x_1 = -2 \quad x_2 = 1 \quad x_3 = -1$

$x = \{-2; 1; -1\}$

d) $x^3 + x^2 = 3x^2 + 3x$

$x^3 - 2x^2 - 3x = 0$

$x(x^2 - 2x - 3) = 0$

$x(x+1)(x-3) = 0$

$x_1 = 0 \quad x_2 = -1 \quad x_3 = 3$

$x^2 - 2x - 3 = (x-3)(x+1)$

$x = \{0; -1; 3\}$

e) $(t^2 - 3)(2 - t^2) = 0$

$(t - \sqrt{3})(t + \sqrt{3})(\sqrt{2} - t)(\sqrt{2} + t) = 0$

$x = \{\pm\sqrt{3}; \pm\sqrt{2}\}$

f) $(x^3 - x)(2x^2 - 1) = 0$

$x(x^2-1)(2x^2-1) = 0$

$x(x-1)(x+1)(\sqrt{2}x - 1)(\sqrt{2}x + 1) = 0$

$\sqrt{2}x = 1$

$x = \frac{1}{\sqrt{2}}$

$x = \frac{\sqrt{2}}{2}$

$x = \{0; 1; -1; \frac{\sqrt{2}}{2}; -\frac{\sqrt{2}}{2}\}$